

1/20 12:30:19 ***



Data Modeling

11e

Database Systems Design, Implementation, and Management



Chapter 2 Data Models

Learning Objectives

- In this chapter, you will learn:
 - About data modeling and why data models are important
 - About the basic data-modeling building blocks
 - What business rules are and how they influence database design

Learning Objectives

- In this chapter, you will learn:
 - How the major data models evolved
 - About emerging alternative data models and the need they fulfill
 - How data models can be classified by their level of abstraction

. "All models are wrong, but some are useful"

Data Modeling and Data Models

- **Data modeling:** Iterative and progressive process of creating a specific data model for a determined problem domain
- **Data models:** Simple representations of complex real-world data structures
- Useful for supporting a specific problem domain
- **Model** - Abstraction of a real-world object or event

Importance of Data Models

Are a communication tool

Give an overall view of the database

Organize data for various users

Are an abstraction for the creation of good database

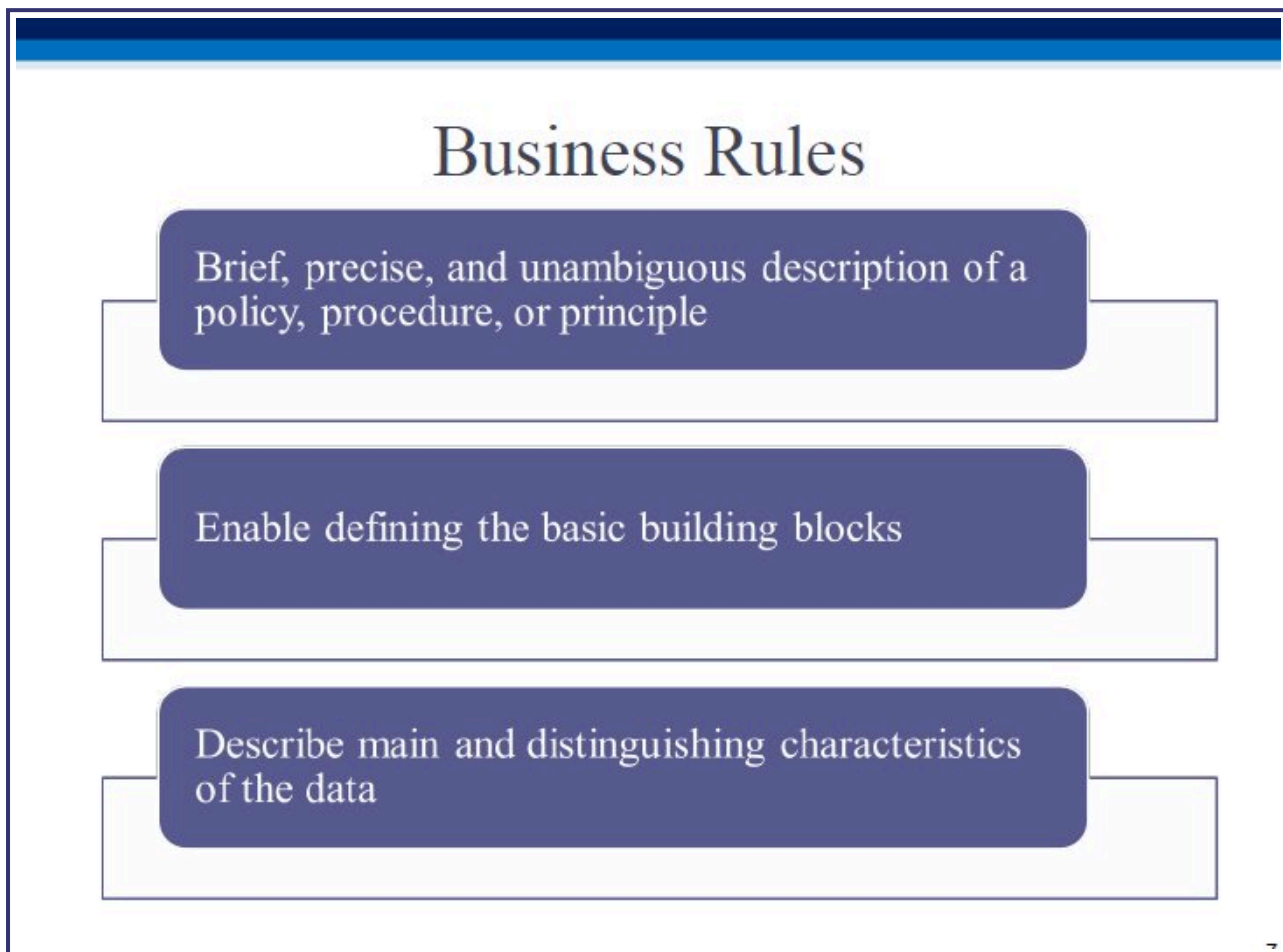
. 3 types of relationships

Data Model Basic Building Blocks

- **Entity:** Unique and distinct object used to collect and store data
 - **Attribute:** Characteristic of an entity
- **Relationship:** Describes an association among entities
 - **One-to-many (1:M)**
 - **Many-to-many (M:N or M:M)**
 - **One-to-one (1:1)**
- **Constraint:** Set of rules to ensure data integrity

Regardless of the model type, TWO entities are relatable in exactly three ways: 1:M, M:N, 1:1.

. 'Business' (DOMAIN) rules: sources of entities and relationships



Sources of Business Rules

Company
managers

Policy makers

Department
managers

Written
documentation

Direct
interviews
with end users

Reasons for Identifying and Documenting Business Rules

- Help standardize company's view of data
- Communications tool between users and designers
- Allow designer to:
 - Understand the nature, role, scope of data, and business processes
 - Develop appropriate relationship participation rules and constraints
 - Create an accurate data model

2

-

Translating Business Rules into Data Model Components

- Nouns translate into entities
- Verbs translate into relationships among entities
- Relationships are bidirectional
- Questions to identify the relationship type
 - How many instances of B are related to one instance of A?
 - How many instances of A are related to one instance of B?

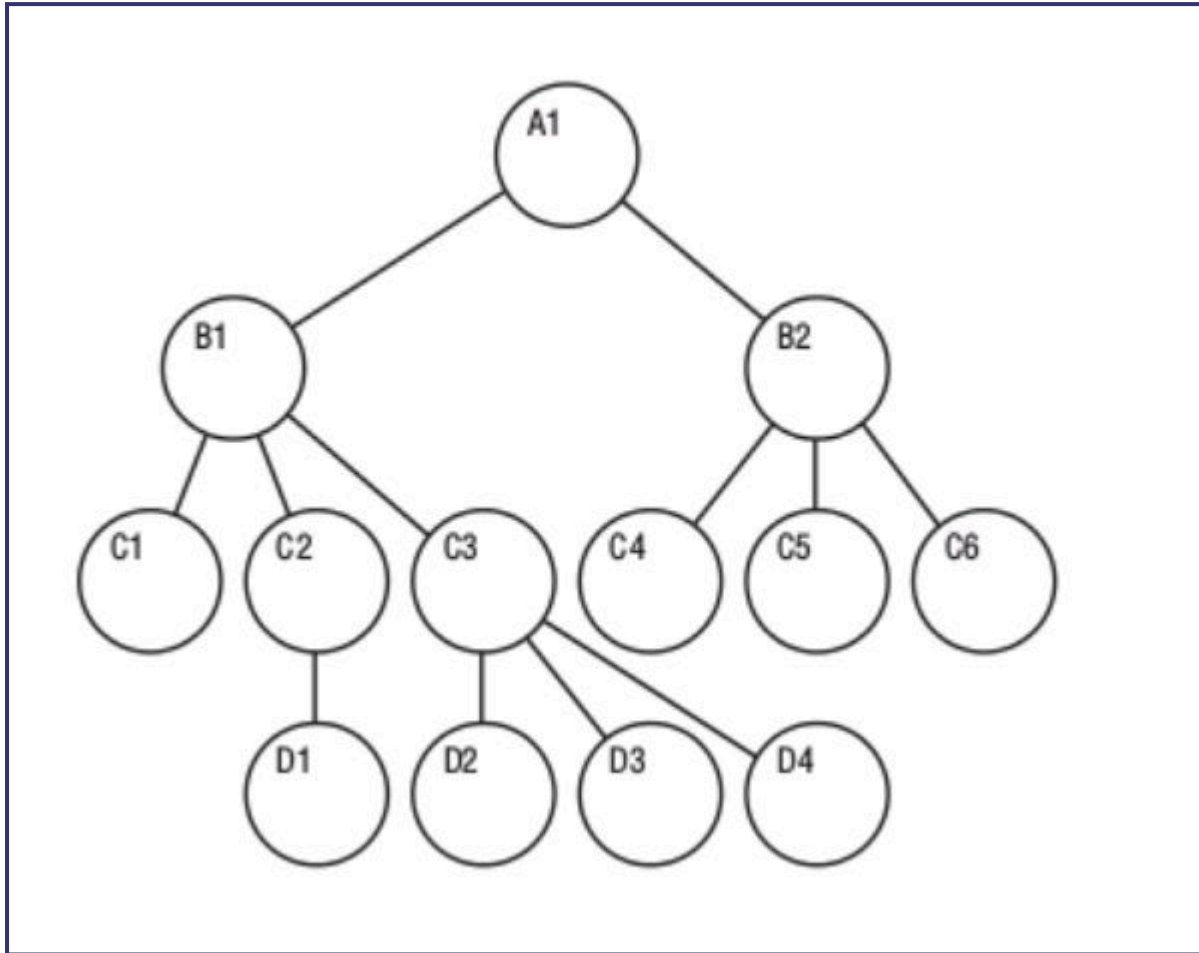
40

Naming Conventions

- Entity names - Required to:
 - Be descriptive of the objects in the business environment
 - Use terminology that is familiar to the users
- Attribute name - Required to be descriptive of the data represented by the attribute
- Proper naming:
 - Facilitates communication between parties
 - Promotes self-documentation

44

Hierarchical modeling



Hierarchical Model

Advantages

- Promotes data sharing
- Parent/child relationship promotes conceptual simplicity and data integrity
- Database security is provided and enforced by DBMS
- Efficient with 1:M relationships

Disadvantages

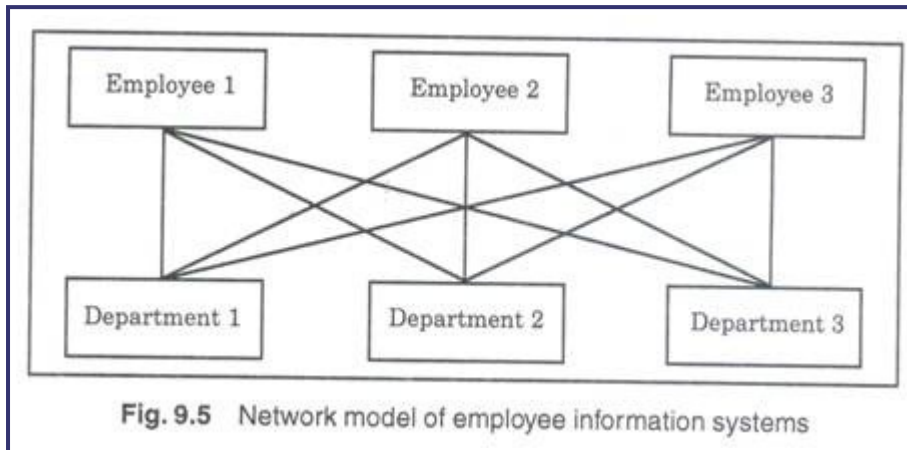
- Requires knowledge of physical data storage characteristics
- Navigational system requires knowledge of hierarchical path
- Changes in structure require changes in all application programs
- Implementation limitations
- No data definition
- Lack of standards

43

At first, data was stored in individual files (transitioned from paper). The next improvement was a 'hierarchical DB model', where data was structured in the form of a tree [similar to a modern filesystem]. Data, in the form of nodes, are linked in a tree-like fashion. To traverse the tree, we need to know the underlying format ('class hierarchy, to make an analogy with classes and objects), and the actual path [eg. to relate A1 and D2, we need to traverse A1->B1->C3>D2].

Hierarchies are good for '1:M' [tree], but not 'M:N' [graph or multiple inheritance].

Network modeling



Network Model

Advantages

- Conceptual simplicity
- Handles more relationship types
- Data access is flexible
- Data owner/member relationship promotes data integrity
- Conformance to standards
- Includes data definition language (DDL) and data manipulation language (DML)

Disadvantages

- System complexity limits efficiency
- Navigational system yields complex implementation, application development, and management
- Structural changes require changes in all application programs

A network model is better than a hierarchical one, because it can capture M:N [in addition to the above, another example is 'products and orders'].

•

Hierarchical and Network Models

Hierarchical Models

- Manage large amounts of data for complex manufacturing projects
- Represented by an upside-down tree which contains segments
 - **Segments:** Equivalent of a file system's record type
- Depicts a set of one-to-many (1:M) relationships

Network Models

- Represent complex data relationships
- Improve database performance and impose a database standard
- Depicts both one-to-many (1:M) and many-to-many (M:N) relationships

13

Data creation, querying

Standard Database Concepts

Schema

- Conceptual organization of the entire database as viewed by the database administrator

Subschema

- Portion of the database seen by the application programs that produce the desired information from the data within the database

Schema data definition language (DDL)

- Enables the database administrator to define the schema components

Data manipulation language (DML)

- Environment in which data can be managed and is used to work with the data in the database

Relational model

The Relational Model

- Based on a relation
 - **Relation** or **table**: Matrix composed of intersecting tuple and attribute
 - **Tuple**: Rows
 - **Attribute**: Columns
- Describes a precise set of data manipulation constructs

Relational Model

Advantages

- Structural independence is promoted using independent tables
- Tabular view improves conceptual simplicity
- Ad hoc query capability is based on SQL
- Isolates the end user from physical-level details
- Improves implementation and management simplicity

Disadvantages

- Requires substantial hardware and system software overhead
- Conceptual simplicity gives untrained people the tools to use a good system poorly
- May promote information problems

Relational Database Management System(RDBMS)

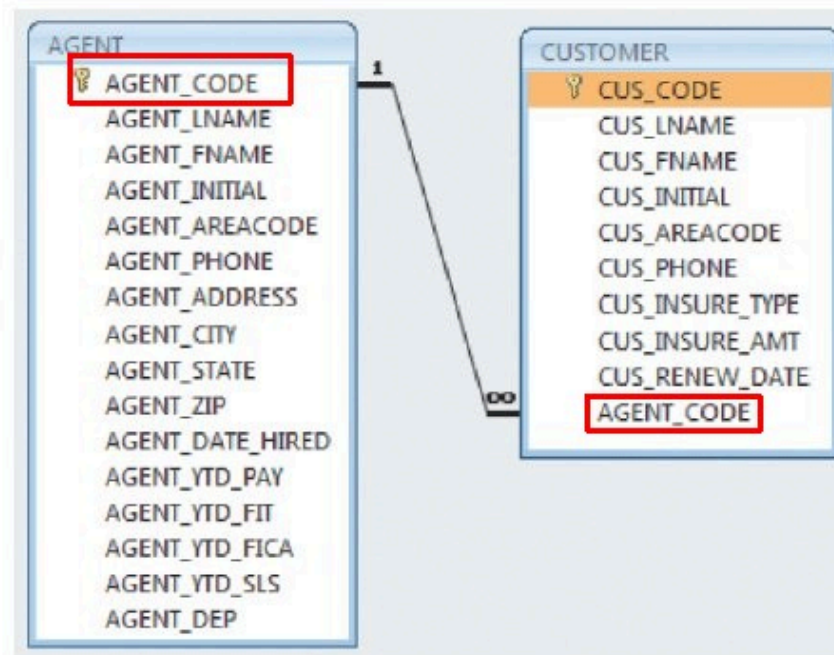
- Performs basic functions provided by the hierarchical and network DBMS systems
- Makes the relational data model easier to understand and implement
- Hides the complexities of the relational model from the user

SQL-Based Relational Database Application

- End-user interface
 - Allows end user to interact with the data
- Collection of tables stored in the database
 - Each table is independent from another
 - Rows in different tables are related based on common values in common attributes
- SQL engine
 - Executes all queries

E-R

Figure 2.2 - A Relational Diagram



Cengage Learning © 2015

The Entity Relationship Model

- Graphical representation of entities and their relationships in a database structure
- **Entity relationship diagram (ERD)**
 - Uses graphic representations to model database components
- **Entity instance or entity occurrence**
 - Rows in the relational table
- **Connectivity**: Term used to label the relationship types

Entity Relationship Model

Advantages

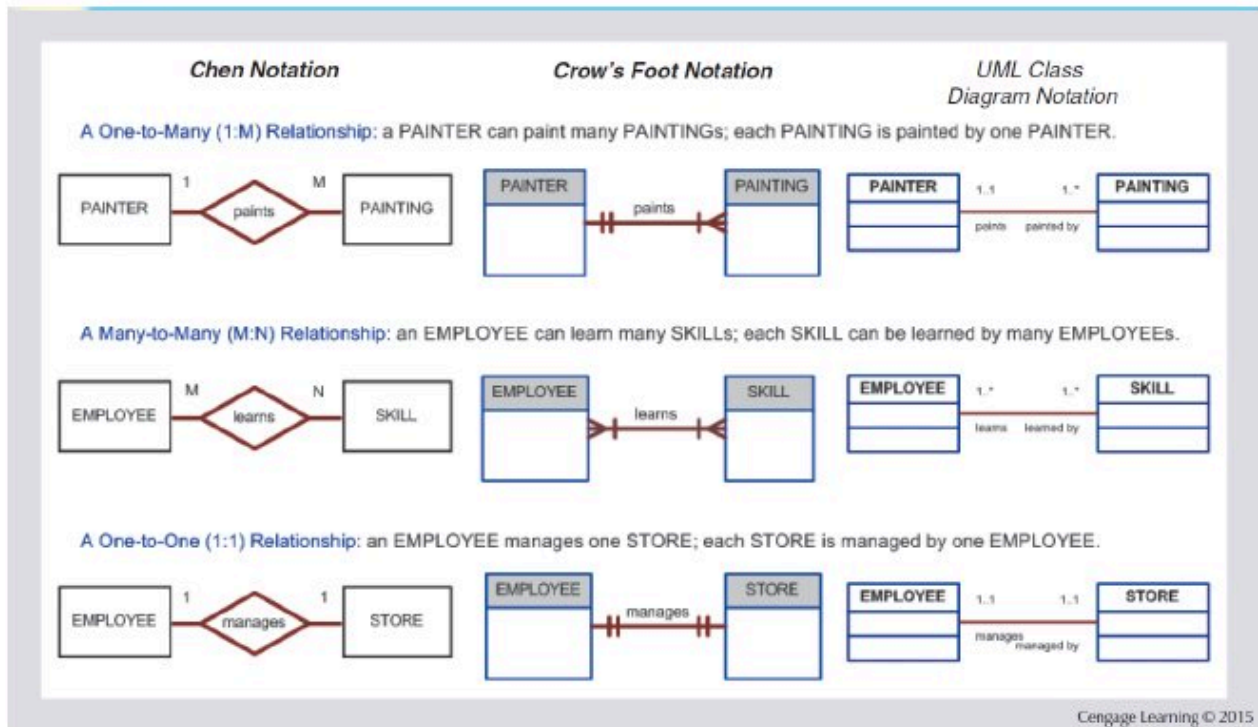
- Visual modeling yields conceptual simplicity
- Visual representation makes it an effective communication tool
- Is integrated with the dominant relational model

Disadvantages

- Limited constraint representation
- Limited relationship representation
- No data manipulation language
- Loss of information content occurs when attributes are removed from entities to avoid crowded displays

Notations

Figure 2.3 - The ER Model Notations



Additional reading: [here is information on, and comparison between, four ER notations: Chen, Crow, Rein85, IDEFIX.](https://www.rein85.com/er-model-notations-comparison/)

Two more models [loosely, 'semantic']: O-O, O-R

O-O DBs: also called 'object stores', these dbs offer(ed) a way to store ("persist") objects on disk. The objects (entity instances) are instanced from classes (entities), like with standard OO programming practice.

Advantages:

- 'cleaner' design - objects mimic real-world counterparts
- inheritance and encapsulation possible
- richer datatypes (attributes) available
- good for CAD, multimedia..

Drawbacks:

- harder to query (compared to relational DBs) - no straightforward way to build and traverse relations between objects
- relations are simpler in certain situations

The RDBMS community collectively ignored this development..

O-R DBs are a compromise between RDBs and OODBs - they feature an O-O front-end over a relational architecture. Interfacing applications do so in an O-O way, and queries/modifications are translated to/from relational form ("ORM").

Benefits:

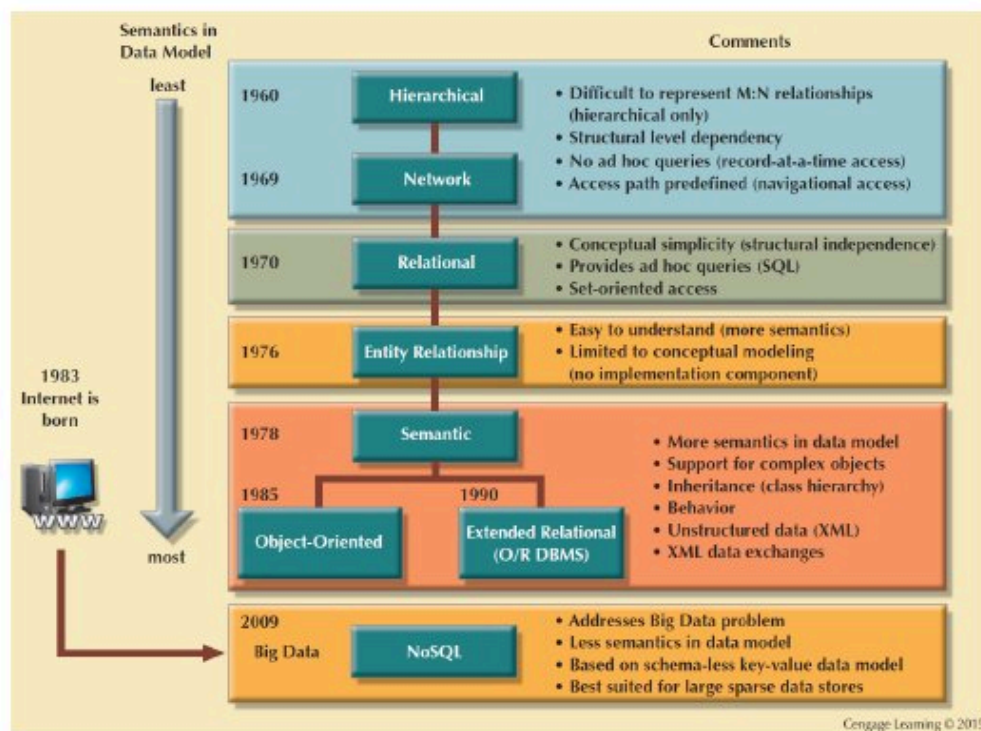
- easy to access the data from an O-O application
- queries can be simpler (can use objects' structure)

Drawback:

- performance can be poor on account of the two-way translation

Data models: hierarchical => => NoSQL => => ...

Figure 2.6 - The Evolution of Data Models

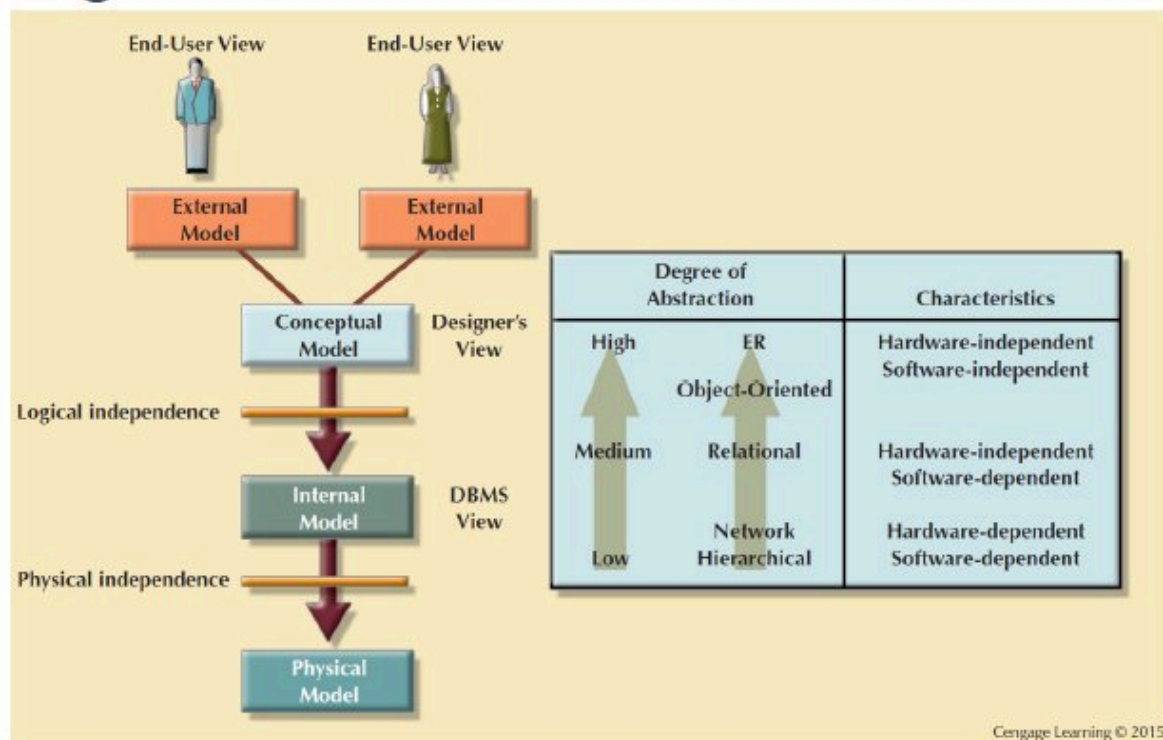


Data models have evolved - from 'hierarchical' (very rigid) to 'NoSQL' (VERY flexible).

The evolution continues... the **vector model** is used to represent data (text, images, audio, video...) as n-D vectors (aka 'embeddings'). Vector DBs (eg. Pinecone and MANY others) store such vectors, and enable their querying (based on SIMILARITY to an input).

Layered data abstraction

Figure 2.7 - Data Abstraction Levels



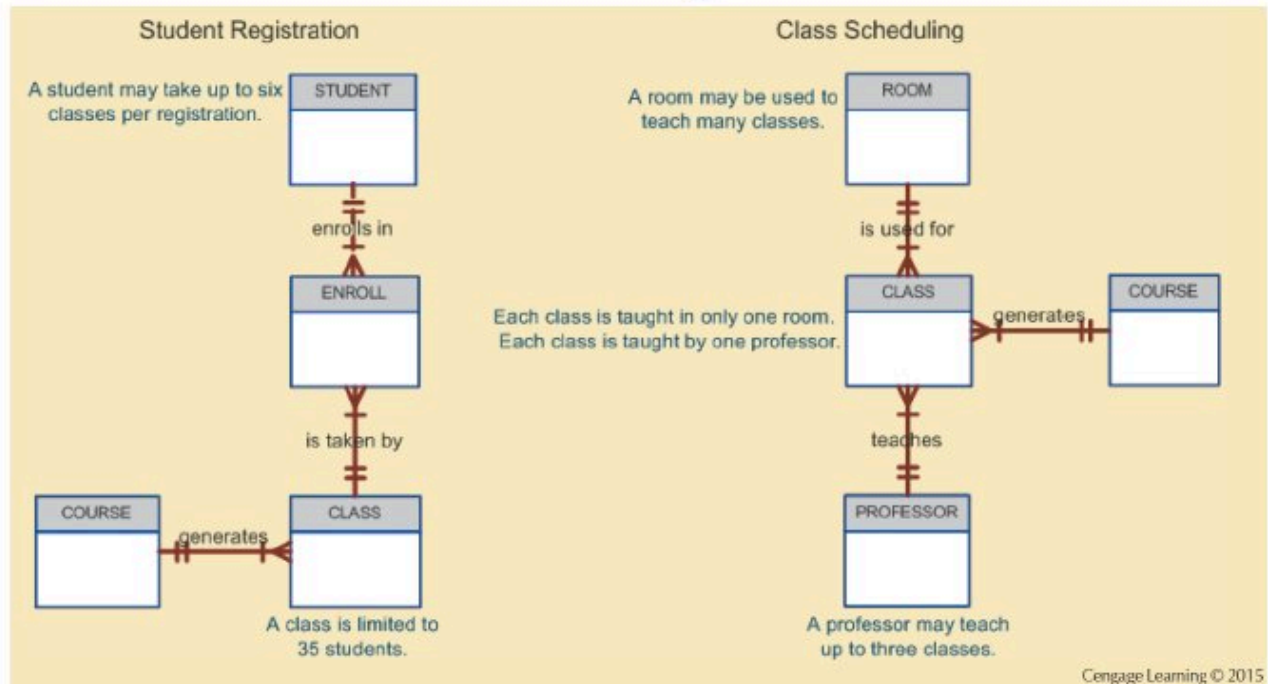
External ['user'] model

The External Model

- End users' view of the data environment
- ER diagrams are used to represent the external views
- **External schema:** Specific representation of an external view

33

Figure 2.8 - External Models for Tiny College



An external model is a collection of 'fragmented', 'from the stakeholders' POV', modeling of a database.

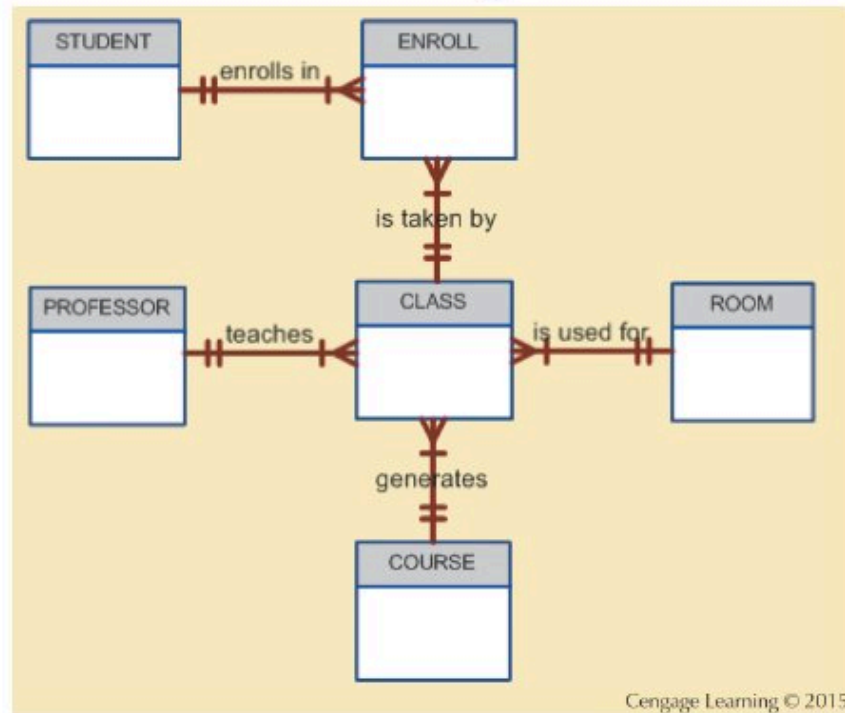
Conceptual ['designer'] model

The Conceptual Model

- Represents a global view of the entire database by the entire organization
- **Conceptual schema:** Basis for the identification and high-level description of the main data objects
- Has a macro-level view of data environment
- Is software and hardware independent
- **Logical design:** Task of creating a conceptual data model

10

Figure 2.9 - Conceptual Model for Tiny College



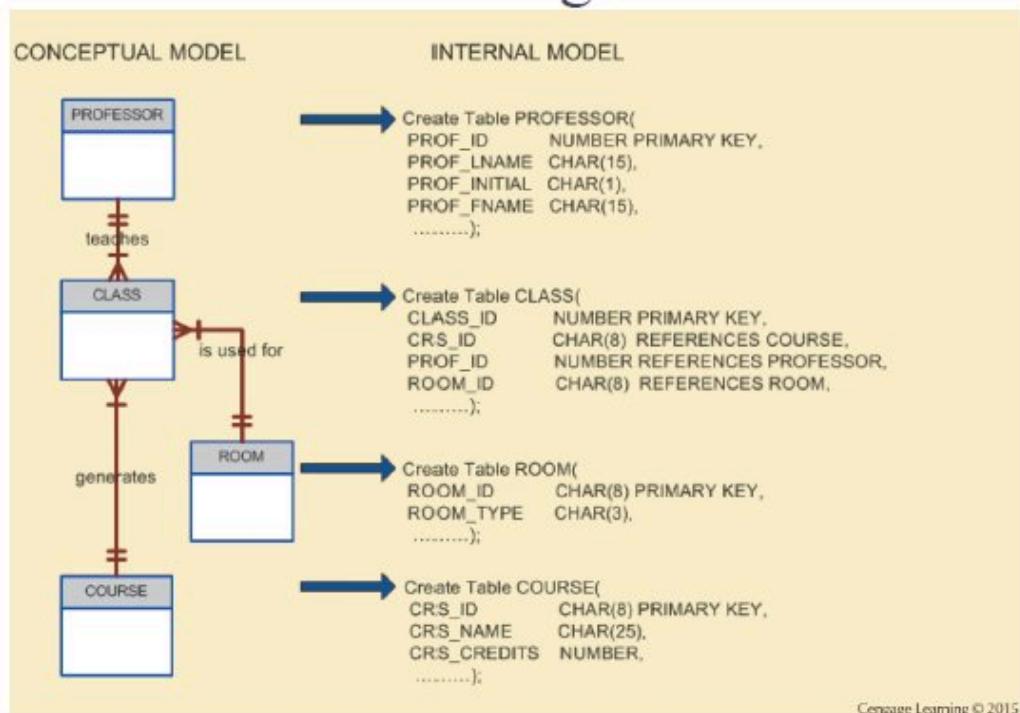
A conceptual model unifies the external views into a cohesive one.

Internal ['programmer'] model

The Internal Model

- Representing database as seen by the DBMS
mapping conceptual model to the DBMS
- **Internal schema:** Specific representation of an internal model
 - Uses the database constructs supported by the chosen database
- Is software dependent and hardware independent
- **Logical independence:** Changing internal model without affecting the conceptual model

Figure 2.10 - Internal Model for Tiny College



Physical ['implementer'] model

The Physical Model

- Operates at lowest level of abstraction
- Describes the way data are saved on storage media such as disks or tapes
- Requires the definition of physical storage and data access methods
- Relational model aimed at logical level
 - Does not require physical-level details
- **Physical independence:** Changes in physical model do not affect internal model

The physical model specifies actual data storage specifics (file format, APIs...).

